TECHNICAL PROJECT REPORT

Coastal Logistics Terminal – Material Sourcing Study

Document prepared as a controlled engineering-document test case for the InfraMiner AI prototype.

1. Project Overview

The proposed Coastal Logistics Terminal includes a marine breakwater, a reclamation platform, internal access roads and concrete infrastructure. The project team must identify suitable material sources while considering material quantity, technical quality, transport distance, availability and supply risk.

2. Breakwater Material Requirement

Estimated rock quantity: The required quantity of rock for breakwater construction is **120,000 tonnes**.

Material: Rock. **Application:** Breakwater construction. **Required quantity:** 120,000 tonnes. The rock shall have high durability and shall be suitable for marine construction works.

3. Other Material Requirements

Material	Required Quantity	Construction Purpose	Requirement
Rock	120,000 tonnes	Breakwater construction	High durability; suitable for marine works
Sand	80,000 m³	Reclamation platform	Clean fill sand
Aggregate	45,000 tonnes	Internal access roads	Suitable road-base quality
Cement	12,000 tonnes	Concrete works	Project-approved grade

4. Candidate Rock Sources

The following candidate sources are provided as controlled project data for evaluating the material-source selection workflow. Values are illustrative and are not live quarry or supplier information.

Source	Material	Distance	Quality	Availability	Risk
North Ridge Quarry	Rock	35 km	92/100	90/100	Low
East Valley Quarry	Rock	48 km	86/100	82/100	Low
Riverbend Quarry	Rock	62 km	90/100	76/100	Medium
South Hill Quarry	Rock	85 km	78/100	88/100	Medium
West Stone Quarry	Rock	110 km	74/100	68/100	High

5. Source Selection Criteria

The prototype decision engine evaluates candidate sources using five weighted criteria. The weights are configurable prototype assumptions and are not presented as an industry standard.

Criterion	Weight	Preferred condition
Cost	30%	Lower cost
Distance	25%	Shorter transport distance
Availability	20%	Higher availability
Quality	15%	Higher technical quality
Risk	10%	Lower supply risk

6. Example Resilience Scenario

If the primary rock source becomes unavailable because of a supply disruption, the decision-support system should remove the unavailable source, recalculate the scores of the remaining sources and recommend the highest-ranked alternative as a backup source.

7. Example Queries for Document Intelligence

Query 1: What material is required for breakwater construction?

Expected answer: Rock, 120,000 tonnes, for breakwater construction.

Query 2: How much sand is required for reclamation?

Expected answer: 80,000 m³ of clean fill sand.

Query 3: What material is required for access roads?

Expected answer: 45,000 tonnes of road-base aggregate.

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